

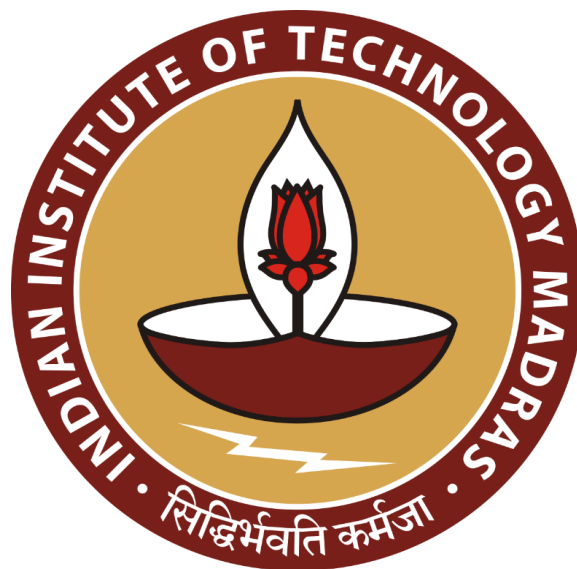
Optimizing Inventory Management and Product Profitability for Aastha RCM PUC

Final Submission for the BDM capstone Project

Submitted by

Name : **Nivedita Prasad**

Roll number : **22F3001078**



IITM Online BS Degree Program,

Indian Institute of Technology, Madras, Chennai

Tamil Nadu, India, 600036

Contents

1	Executive Summary	3
2	Analysis Processes and Methods	4
3	Results, Findings and Interpretation of Results	8
4	Recommendations	21

1 Executive Summary

This report presents a data-driven strategy to enhance inventory management and product profitability for Aastha RCM PUC, a department store facing challenges related to inventory overstock, stockouts, and fluctuating demand. The primary objective was to identify actionable insights to minimize excess inventory, reduce stockouts, and maximize profitability by leveraging business data management techniques.

The analysis began with comprehensive data collection from the store's point-of-sale (POS) system, covering 21 weeks of sales and inventory data. Data cleaning and organization were followed by systematic analysis using key metrics such as the Coefficient of Variation (CV%) for demand fluctuation, Simple Moving Average (SMA) for demand forecasting, and ABC analysis for inventory prioritization. Product profitability was evaluated by examining total profit and sales volume across all categories.

Key findings reveal that most products (75%) experience moderate demand fluctuations, with only a small proportion showing stable (22%) or highly variable (3%) demand. Demand forecasting using SMA highlighted consistent growth trends in essential grocery items, health supplements, and personal care products. Inventory optimization analysis showed that overstock is a prevalent issue across all product categories, especially for lower-value items, while optimal stock levels remain consistently lower than overstock and stockout figures. Product profitability analysis identified top performers such as Rice Bran Oil and Detergent Cake, which contributed significantly to both profit and sales volume. Based on these insights, the report recommends tailored inventory strategies for products with different demand patterns, prioritizing high-value items, reducing overstock in low-value categories, and promoting high-margin, high-volume products. Additionally, it suggests reviewing and potentially discontinuing low-performing products and optimizing pricing and promotions to improve overall business performance. Overall, this report provides a clear approach for Aastha RCM PUC to achieve more efficient inventory management and enhanced profitability.

2 Analysis Processes and Methods

Data Collection and Preparation:

The department store Aastha RCM PUC has POS(point of sale), a computer-based system that allows businesses to keep track of sales and inventory data of each product of the store. The owner, Mrs. Sarika Devi provided me the username and password of the POS dashboard. So, the excel files having compiled report of weekly data which includes comprehensive records related to inventory movement, sales, and stock status were downloaded for the time period of 21 weeks from May 2024-September 2024 directly from the system.

Then, the data cleaning was performed by deleting the rows where most of the column values were zero, those are considered non-informative for the analysis. Some columns like tax rate were also deleted. Then, we organized the data into separate sheets for calculation and analysis and renamed columns for clarity.

Demand Fluctuation Analysis:

The objective of this analysis was to categorize products based on their demand stability and fluctuation levels over time. This was done using the Coefficient of Variation (CV%), a measure that standardizes demand fluctuations relative to the average sales.

For each product, weekly sales data (Quantity Out) was used as the basis for analysis. Sales values were arranged in a format where each product's sales were tracked over 21 weeks. Coefficient of Variation (CV%) was used as the primary metric. To compute CV%, Average Sales and Standard Deviation were calculated for each product over 21 weeks, and then CV% was calculated as

$$CV\% = (\text{Standard deviation}/\text{Mean}) * 100$$

Here, the Standard Deviation measures the amount of variation or dispersion in sales data, and the Mean is the average sales value.

This metric provides a normalized measure of variability relative to the average demand, allowing for fair comparisons across products with different sales volumes.

Then, the products were classified into three demand fluctuation categories:

Stable Demand: $CV\% \leq 50\%$

Moderate Fluctuation: $50\% < CV\% \leq 100\%$

High Fluctuation: $CV\% > 100\%$

CV% based analysis enables businesses to design inventory strategies based on product demand behavior. Stable-demand items can be managed with fixed reorder points and minimal safety stock,

while fluctuating-demand items require dynamic stock levels and regular review. This approach reduces overstock and stockout risks and thus enhances inventory optimization.

A classification formula was applied in Excel using IF() logic to assign categories to each product.

Then, a summary table was created showing the count of products in each category and a pie chart was generated to represent the proportional distribution of products by demand type. We are using a pie chart to represent the distribution of products based on demand fluctuation categories because it is ideal for visualizing proportional relationships in a dataset than any other graph.

Demand Forecasting with Moving Average:

This analysis was done to understand weekly sales trends for different categories of products and to forecast near-future demand. The Simple Moving Average (SMA) method was selected to achieve this. The SMA is computed by taking the average of the actual sales(Qty Out) over a fixed number of past weeks. This average serves as the forecast for the next week. We calculated SMA by taking the average sales quantity of the previous three weeks for each product, starting from Week 4, and extending it up to Week 22, as follows

$$\text{SMA(Week 22)} = (\text{Sales(Week 19)} + \text{Sales(Week 20)} + \text{Sales(Week 21)}) / 3$$

For week 23, the SMA was calculated using actual sales of weeks 20, 21, and the forecasted value of week 22. Similarly, for week 24, the average was taken of week 21 sales, week 22 forecast, and week 23 forecast, as follows

$$\text{SMA(Week 23)} = (\text{Sales(Week 20)} + \text{Sales(Week 21)} + \text{SMA(Week 22)}) / 3$$

$$\text{SMA(Week 24)} = (\text{Sales(Week 21)} + \text{SMA(Week 22)} + \text{SMA(Week 23)}) / 3$$

The Simple Moving Average method was chosen because it allows for quick and practical insights into a realistic trend line that could be used to forecast short-term demand and make better inventory and ordering decisions without requiring complex statistical techniques. Forecasting future demand helps predict how much stock will be needed in the upcoming weeks. By accurately projecting demand for each product, stockouts can be minimized because the business can order or stock up in advance based on forecasted sales, especially for high-demand products. Also, overstock situations are reduced, as forecasting avoids unnecessary inventory buildup for low-demand items, which can tie up capital and storage space.

We are using line charts for visualization because they clearly show trends over time, patterns, fluctuations, and direction of change at a glance. When we overlay the actual sales and the simple moving average (SMA) on the same chart, we can visually compare real behavior with the forecasted trend. This helps us identify whether sales are stable, increasing, decreasing, or irregular.

Inventory Optimization:

To begin the inventory analysis for Aastha RCM PUC, we first tabulated the weekly sales data for each product over the 21 weeks. This allowed us to compute both the average weekly usage and the maximum weekly usage, which were essential inputs for calculating safety stock, ROP, and maximum threshold. Using these parameters, we calculated the Safety Stock for each product using the formula:

$$\text{Safety Stock} = \text{Max Weekly Usage} - \text{Average Weekly Usage}$$

This safety stock is the extra inventory kept on hand to protect against uncertainties in demand or supply. Next, we calculated the Reorder Point(ROP) using the formula:

$$\text{ROP} = \text{Demand during Lead Time} + \text{Safety Stock}$$

$$\text{or, ROP} = (\text{Average Weekly Usage} \times \text{Lead Time}) + \text{Safety Stock}$$

This helped us determine the inventory level at which a new order should be placed to replenish stock before it runs out, factoring in both expected demand during lead time and safety stock.

We then calculated the Maximum Threshold, representing the upper stock limit as

$$\text{Maximum threshold} = \text{Max Weekly Usage} + \text{Safety Stock}$$

We defined a Maximum Threshold as a control limit, above which excess inventory would be considered overstock.

Following this, we conducted an ABC Analysis to classify inventory items based on their contribution to total sales value. For this, we calculated the total sales value over the 21 weeks from the sell amount data, ranked the products in descending order of value, and computed both the Cumulative Total and Cumulative Percentage of total consumption. Based on the thresholds A = top 80%, B = next 15%, C = bottom 5%, items were categorized into A, B, or C classes. This classification follows the Pareto Principle (80/20 rule), focusing on prioritizing the most valuable items.

We categorized stock levels into three groups using the following formula in Excel:

$$=IF(B2 < C2, "Stockout", IF(B2 > E2, "Overstock", "Optimal"))$$

Where, B2 is Balance, which is the closing stock of Week 21, C2 is the Reorder Point, and E2 is the Maximum Threshold.

This determines the stock status based on conditions:

If the Balance is less than the Reorder Point, it indicates a stockout.

If the Balance in Week 21 is greater than the Maximum Threshold, it indicates overstock.

Otherwise, the stock level is considered optimal.

Then, we summarized the number of units falling under Stockout, Optimal, and Overstock status for each ABC category. These totals were computed and then visualized using a clustered bar chart, helping

identify which categories have excessive or insufficient stock and where to focus stock optimization efforts.

Product Profitability Analysis:

To understand product-level performance at Aastha RCM PUC, we analyzed two key metrics, Total Profit and Total Sales Volume, using Microsoft Excel. The objective was to identify the high-margin products, high-volume products, and low-performing or underutilized items. For this, we first tabulated the profit and sales volume of each product for each week of the 21 weeks in two different sheets. Then, Excel's Sort function was used to rank products in descending order, both by Total Profit and then by Total Sales Volume. This sorting helped us focus on the top ten in each category.

A Pie Chart was then created to show the share of each category in the overall profit. To do this, we first aggregated the total profit for each product category by summing the profits of all products within a category. Then, we used Excel's Insert Pie Chart feature and customized it by adding data labels as percentages to visually represent each category's contribution to the total profit.

Further, we tabulated the total profit and total sales volume for each product across the 21-week period. Then, conditional formatting in Excel was applied to create a Heatmap using a three-color scale, which is red for low sales or profit values, yellow for mid-range values, and green for high sales or profit values. This color-coded visualization made it easier to mark outliers such as products with high sales but low profit, or low sales but high profit. The combined use of mathematical calculation, sorting, and visualization allowed for a comprehensive and actionable product performance analysis.

3 Results, Findings, and Interpretations of Results

Demand Fluctuation Analysis:

The analysis categorizes 36 products based on their demand fluctuation, measured by the coefficient of variation (CV%).

Products are grouped into three categories:

- (i) Stable Demand: 8 products (22%)
- (ii) Moderate Fluctuation: 27 products (75%)
- (iii) High Fluctuation: 1 product (3%)

A significant portion, accounting for 22% of the total, exhibits stable demand, meaning their sales are consistent and predictable over time. The majority of products (75%) fall under the Moderate Fluctuation category, indicating that most products experience some variability in demand but not extreme fluctuation. Only a small proportion of products (3%) show High Fluctuation in demand, suggesting that highly unpredictable demand is rare among the analyzed items.

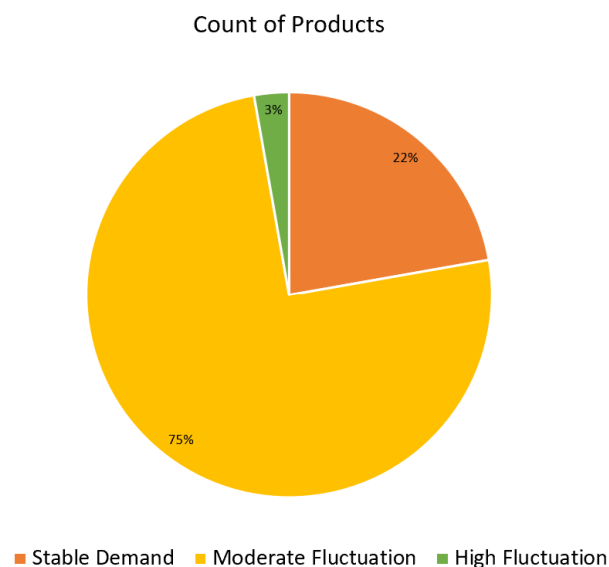


Figure 3.1: Distribution of products based on demand fluctuation

The pie chart visually illustrates these proportions, with the largest segment representing moderate fluctuation, followed by stable demand, and a very small segment for high fluctuation.

Demand Forecasting with Moving Average:

Demand forecasting was done for all the categories of products which are Grocery, Food & Health Supplements, Personal Care & Home Care products, and Apparel & Footwear.

Demand Forecasting for Week 22-24 for the products of Grocery category:

The last three SMA values are the demand forecasting for the week 22, week 23 and week 24.

Week no.	Grocery									
	Masoor Malka(Qty Out)	Masoor Malka(SMA)	Tea(Qty Out)	Tea(SMA)	Chilli Powder(Qty Out)	Chilli Powder(SMA)	Coriander Powder(Qty Out)	Coriander Powder(SMA)	Turmeric Powder(Qty Out)	Turmeric Powder(SMA)
1	9	-	24	-	6	-	7	-	32	-
2	18	-	34	-	15	-	18	-	47	-
3	12	-	44	-	18	-	13	-	76	-
4	35	13	46	34	35	13	37	13	69	52
5	35	22	50	41	32	23	32	23	81	64
6	14	27	49	47	4	28	14	27	80	75
7	13	28	52	48	3	24	14	28	85	77
8	26	21	76	50	18	13	21	20	98	82
9	40	18	108	59	22	8	31	16	129	88
10	29	26	82	79	6	14	21	22	108	104
11	49	32	117	89	37	15	41	24	151	112
12	63	39	123	102	39	22	56	31	154	129
13	73	47	121	107	97	27	67	39	150	138
14	53	62	162	120	30	58	43	55	192	152
15	54	63	117	135	52	55	45	55	153	165
16	72	60	220	133	52	60	60	52	263	165
17	85	60	161	166	64	45	110	49	214	203
18	97	70	171	166	58	56	90	72	225	210
19	91	85	184	184	46	58	93	87	226	234
20	117	91	230	172	72	56	123	98	264	222
21	112	102	214	195	40	59	118	102	252	238
22		107		209		53		111		247
23		112		218		55		117		254
24		110		214		49		116		251

Table 3.1: Actual Sales and Calculated SMA of different products of Grocery category

Garam Masala(Qty Out)	Garam Masala(SMA)	Sabji Masala(Qty Out)	Sabji Masala(SMA)	Atta(Qty Out)	Atta(SMA)	Rice Bran Oil(Qty Out)	Rice Bran Oil(SMA)	Salt(Qty Out)	Salt(SMA)
16	-	44	-	17	-	219	-	134	-
25	-	59	-	26	-	270	-	225	-
33	-	79	-	36	-	365	-	218	-
51	25	110	61	23	26	415	285	266	192
59	36	152	83	24	28	591	350	314	236
33	48	137	114	24	28	588	457	340	266
33	48	146	133	24	24	615	531	367	307
34	42	169	145	32	24	705	598	440	340
51	33	189	151	62	27	802	636	614	382
37	39	190	168	35	39	854	707	690	474
69	41	216	183	69	43	931	787	572	581
74	52	270	198	70	55	982	862	885	625
132	60	267	225	64	58	1037	922	690	716
65	92	250	251	106	68	1064	983	643	716
89	90	257	262	67	80	1142	1028	725	739
89	95	279	258	174	79	1200	1081	712	686
106	81	316	262	103	116	1387	1135	809	693
101	95	329	284	122	115	1522	1243	825	749
94	99	341	308	112	133	1456	1370	864	782
125	100	390	329	135	112	1671	1455	972	833
92	107	398	353	124	123	1718	1550	1043	887
	104		376		124		1615		960
	107		388		128		1668		992
	101		387		125		1667		998

Table 3.2: Actual Sales and Calculated SMA of different products of Grocery category

Line Charts of Actual Sales vs. Simple Moving Average for Grocery Category Products:

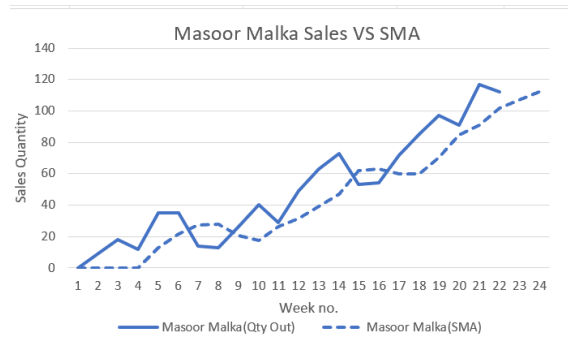


Figure 3.2: Masoor Malka Actual Sales VS SMA

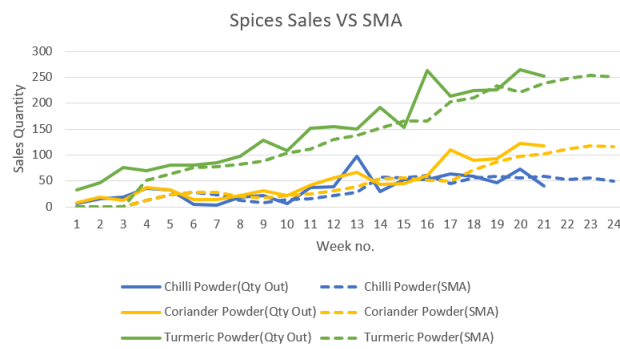


Figure 3.3: Spices Actual Sales VS SMA

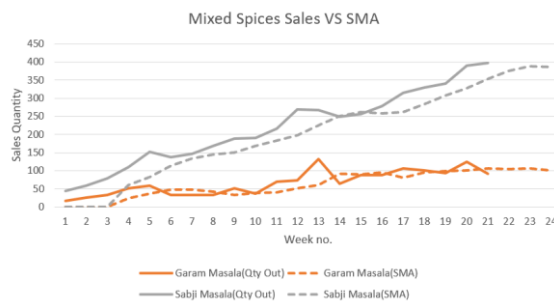


Figure 3.4: Mixed Spices Actual Sales VS SMA

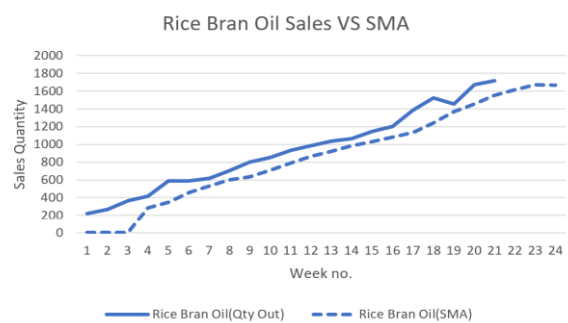
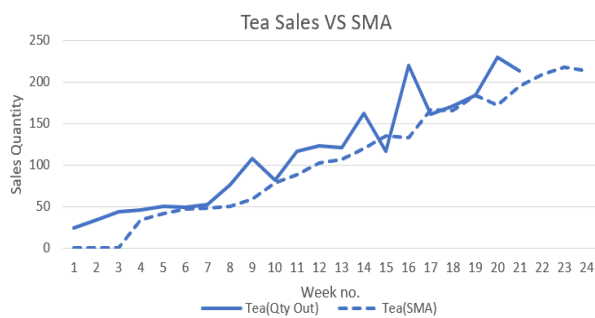


Figure 3.5: Tea Actual Sales VS SMA

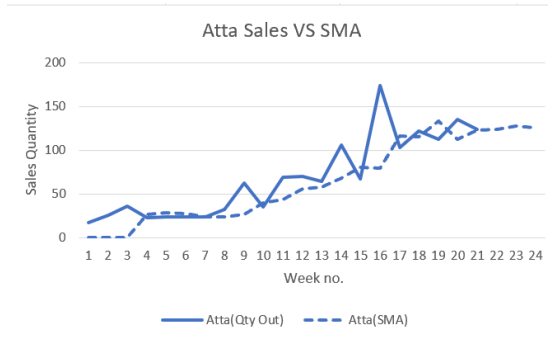


Figure 3.7: Atta Actual Sales VS SMA

Figure 3.6: Rice Bran Oil Actual Sales VS SMA

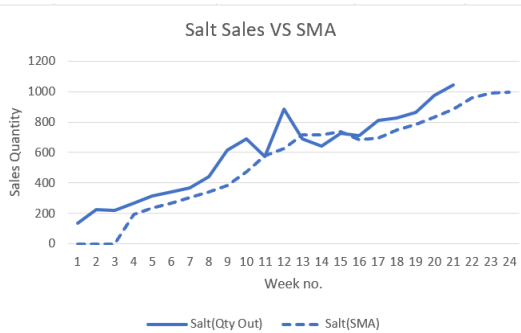


Figure 3.8: Salt Actual Sales VS SMA

The grocery category shows a clear upward sales trend across everyday items. Products like Masoor Malka, Tea, and Rice Bran Oil continue to grow steadily, with Rice Bran Oil showing especially strong growth. Atta and Salt also see rising trends, with small ups and downs. Among spices, Chilli Powder shows steady increases, while Coriander and Turmeric Powders have more noticeable shifts. Mixed spices like Garam Masala maintain steady sales, but Sabji Masala is growing quickly and appears to be in high demand. The simple moving average (SMA) lines help smooth out short-term fluctuations and make it easier to see the overall direction of customer demand over time.

Demand Forecasting for Week 22-24 for the products of Food and Health Supplement categories:

The last three SMA values are the demand forecasting for the week 22, week 23 and week 24.

Week	Noodles		Fiber Biscuit		Jeera Biscuit		Food and Health Supplement		Mixed Pickle		Ketchup	
	(Qty Out)	(SMA)	(Qty Out)	(SMA)	(Qty Out)	(SMA)	(Qty Out)	(SMA)	(Qty Out)	(SMA)	(Qty Out)	(SMA)
1	82	-	13	-	19	-	9	-	11	-	-	-
2	105	-	21	-	32	-	18	-	19	-	-	-
3	126	-	32	-	34	-	10	-	29	-	-	-
4	160	104	20	22	58	28	35	12	17	20	-	-
5	203	130	21	24	61	41	28	21	15	22	-	-
6	208	163	22	24	36	51	7	24	14	20	-	-
7	343	190	28	21	42	52	6	23	18	15	-	-
8	310	251	40	24	61	46	11	14	19	16	-	-
9	360	287	67	30	82	46	18	8	46	17	-	-
10	377	338	49	45	68	62	7	12	21	28	-	-
11	419	349	82	52	115	70	26	12	54	29	-	-
12	555	385	81	66	127	88	41	17	54	40	-	-
13	519	450	75	71	151	103	52	25	44	43	-	-
14	506	498	120	79	117	131	28	40	86	51	-	-
15	516	527	75	92	139	132	29	40	40	61	-	-
16	557	514	186	90	143	136	44	36	144	57	-	-
17	664	526	126	127	175	133	55	34	71	90	-	-
18	727	579	137	129	171	152	65	43	79	85	-	-
19	749	649	144	150	167	163	52	55	68	98	-	-
20	892	713	165	136	202	171	71	57	94	73	-	-
21	928	789	151	149	170	180	67	63	74	80	-	-
22		856		153		180		63		79	-	-
23		892		156		184		67		82	-	-
24		892		154		178		66		78	-	-

Table 3.3: Actual Sales and Calculated SMA of different products of Food and Health Supplement category

Gooddot Proteiz(Qty Out)	Gooddot Proteiz(SMA)	Nutricharge Woman(Qty Out)	Nutricharge Woman(SMA)	Nutricharge Veg Omega(Qty Out)	Nutricharge Veg Omega(SMA)	Nutricharge Man(Qty Out)	Nutricharge Man(SMA)	Mixture(Qty Out)	Mixture(SMA)	Bhujia(Qty Out)	Bhujia(SMA)
16	-	27	-	13	-	17	-	8	-	14	-
27	-	38	-	22	-	28	-	18	-	24	-
34	-	36	-	32	-	35	-	11	-	36	-
52	26	63	34	20	22	55	27	37	12	38	25
53	38	65	46	26	25	57	39	32	22	52	33
28	46	49	55	25	26	30	49	14	27	58	42
28	44	55	59	26	24	35	47	13	28	62	49
37	36	70	56	33	26	42	41	20	20	74	57
57	31	83	58	61	28	62	36	26	16	104	65
51	41	83	69	37	40	52	46	20	20	84	80
83	48	104	79	70	44	85	52	41	22	116	87
89	64	121	90	70	56	90	66	56	29	127	101
149	74	139	103	61	59	151	76	68	39	117	109
86	107	117	121	102	67	86	109	44	55	163	120
111	108	123	126	57	78	113	109	46	56	117	136
115	115	301	126	163	73	119	117	59	53	223	132
138	104	163	180	89	107	137	106	70	50	216	168
148	121	181	196	99	103	134	123	83	58	227	185
144	134	175	215	88	117	122	130	71	71	235	222
180	143	197	173	108	92	143	131	93	75	260	226
147	157	196	184	90	98	114	133	90	82	259	241
	157		189		95		126		85		251
	161		194		98		128		89		257
	155		193		94		123		88		256

Table 3.4: Actual Sales and Calculated SMA of different products of Food and Health Supplement category

Line Charts of Actual Sales vs. Simple Moving Average for Food and Health Supplement Categories of Products:

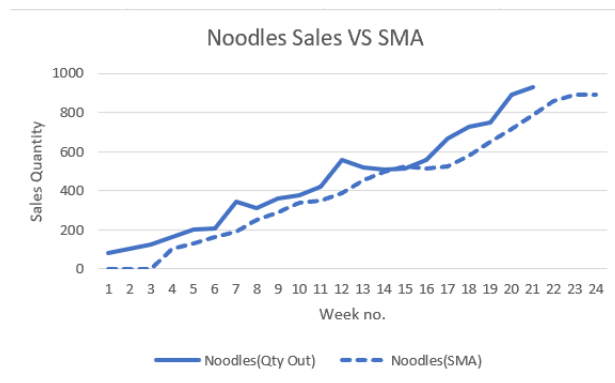


Figure 3.9: Noodles Sales VS SMA

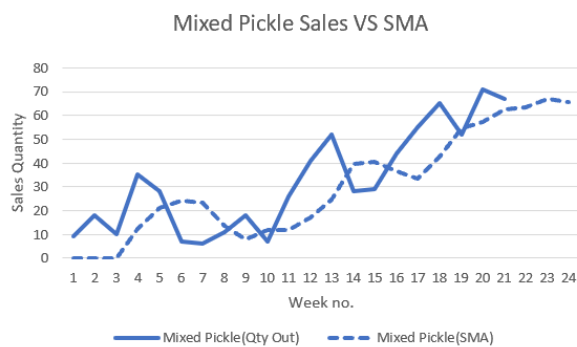


Figure 3.10: Mixed Pickle Sales VS SMA

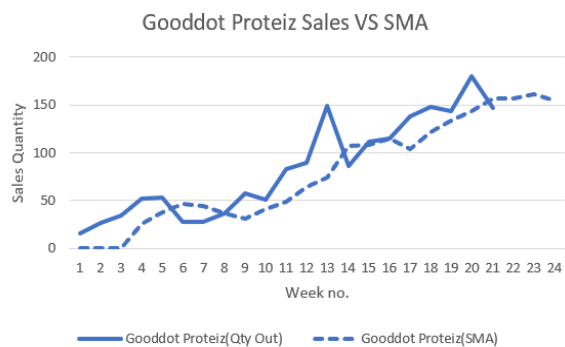


Figure 3.11: Gooddot Proteiz Sales VS SMA

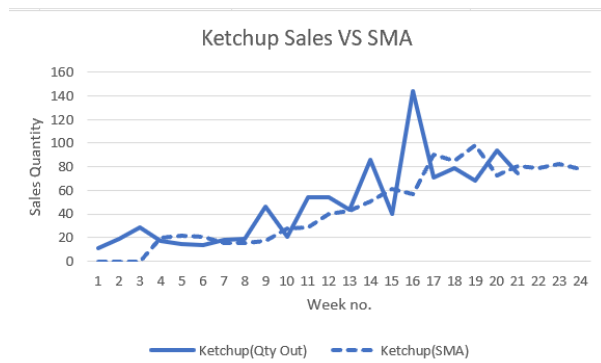


Figure 3.12: Ketchup Sales VS SMA

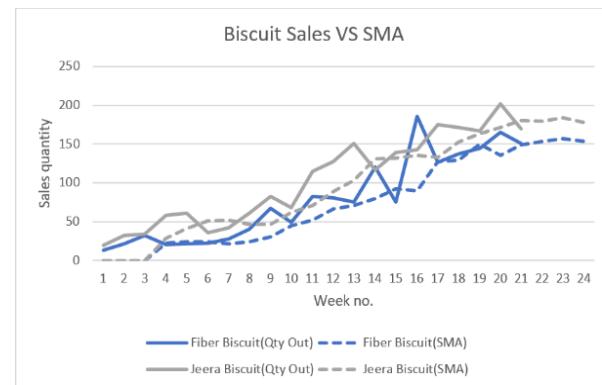


Figure 3.13: Biscuit Sales VS SMA

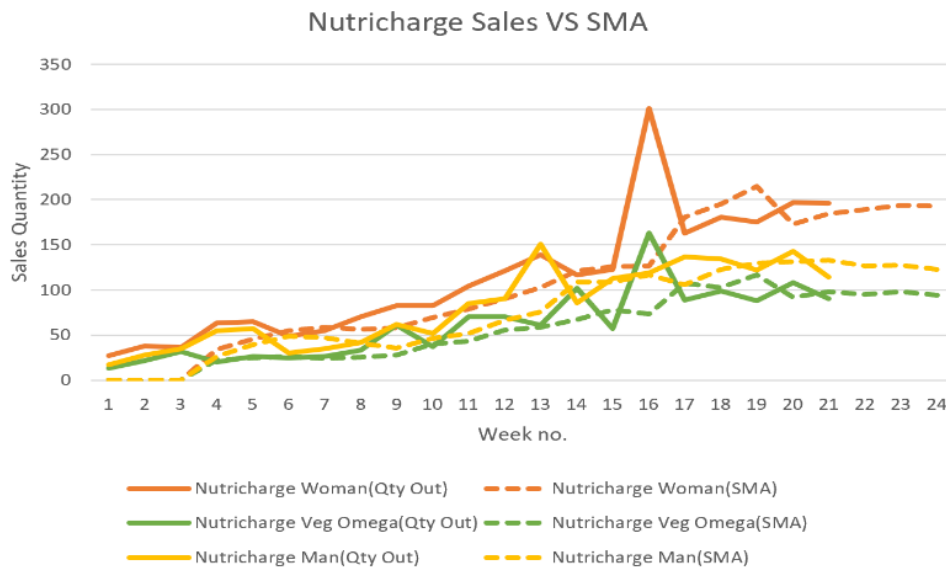


Figure 3.14: Actual Sales VS SMA of Nutrcharge products

This category also shows increasing customer interest, both in easy-to-use and health-related products. Noodles have a steady, straight-line growth, showing consistent popularity. Biscuits, especially Jeera and Snacks like Bhujia are also growing steadily, pointing to their popularity as everyday snack options. Ketchup and Mixed Pickle have uneven trends with noticeable spikes. Meanwhile, Gooddot Proteiz and Nutrcharge items are becoming more popular, with occasional sharp rises especially in the case of Nutrcharge Woman. Overall, the SMA curves help clarify that even with some short-term fluctuations, most food products are becoming more popular over time, showing a shift in customer choices toward both convenience and healthier options.

Demand Forecasting for Week 22-24 for the products of Personal Care and Home Care categories:

Week	Personal Care and Home Care Products							
	Herbal Tooth Paste(Qty Out)	Herbal Tooth Paste(SMA)	Multi Purpose Cream(Qty Out)	Multi Purpose Cream(SMA)	Shampoo(Qty Out)	Shampoo(SMA)	All Rounder Tooth Paste(Qty Out)	All Rounder Tooth Paste(SMA)
1	9	-	31	-	8	-	38	-
2	81	-	40	-	17	-	55	-
3	14	-	62	-	20	-	146	-
4	41	35	53	44	40	15	178	80
5	34	45	57	52	40	26	185	126
6	14	30	65	57	12	33	185	170
7	13	30	66	58	11	31	201	183
8	36	20	66	63	15	21	229	190
9	43	21	93	66	34	13	246	205
10	33	31	94	75	17	20	238	225
11	55	37	126	84	47	22	268	238
12	69	44	133	104	49	33	283	251
13	79	52	122	118	106	38	297	263
14	56	68	163	127	39	67	273	283
15	58	68	117	139	61	65	275	284
16	72	64	234	134	62	69	314	282
17	93	62	171	171	75	54	386	287
18	106	74	184	174	70	66	401	325
19	114	90	193	196	57	69	445	367
20	140	104	221	183	77	67	420	411
21	136	120	211	199	43	68	517	422
22		130		208		59		461
23		135		213		60		466
24		134		211		54		481

Table 3.5: Actual Sales and Calculated SMA of different products of Personal Care and Home Care categories

Detergent Cake(Qty Out)	Detergent Cake(SMA)	Dishwash Bar(Qty Out)	Dishwash Bar(SMA)	Anti-Bacterial Soap(Qty Out)	Anti-Bacterial Soap(SMA)	Jasmine Bath Soap(Qty Out)	Jasmine Bath Soap(SMA)	Washing Powder(Qty Out)	Washing Powder(SMA)
291	-	167	-	61	-	85	-	68	-
381	-	235	-	74	-	108	-	103	-
516	-	261	-	74	-	122	-	122	-
596	396	331	221	110	70	120	105	158	98
812	498	382	276	120	86	158	117	198	128
906	641	400	325	120	101	161	133	187	159
1091	771	445	371	130	117	177	146	200	181
1346	936	516	409	151	123	205	165	236	195
1511	1114	600	454	171	134	264	181	285	208
1520	1316	612	520	167	151	251	215	281	240
1685	1459	674	576	201	163	292	240	323	267
1741	1572	695	629	230	180	299	269	337	296
1783	1649	831	660	248	199	314	281	408	314
1906	1736	782	733	230	226	361	302	346	356
1873	1810	815	769	235	236	323	325	371	364
2020	1854	845	809	253	238	441	333	383	375
2325	1933	1007	814	281	239	401	375	444	367
2437	2073	1048	889	300	256	416	388	450	399
2717	2261	1117	967	300	278	433	419	461	426
2813	2493	1237	1057	338	294	428	417	511	452
3033	2656	1289	1134	342	313	490	426	494	474
	2854		1214		327		450		489
	2900		1247		336		456		498
	2929		1250		335		465		494

Table 3.6: Actual Sales and Calculated SMA of different products of Personal Care and Home Care categories

Line Charts of Actual Sales vs. Simple Moving Average for Personal Care and Home Care Categories of Products:

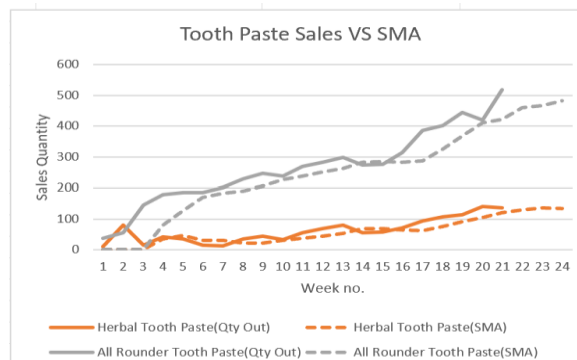


Figure 3.15: Toothpaste Sales VS SMA

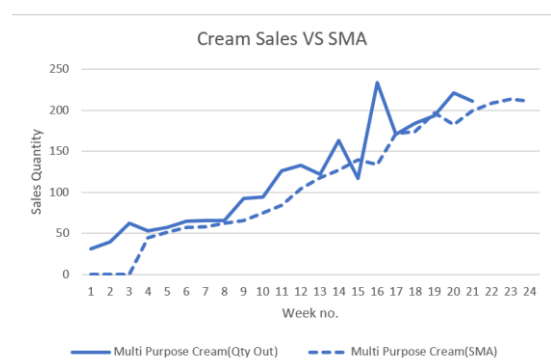


Figure 3.16: Cream Sales VS SMA

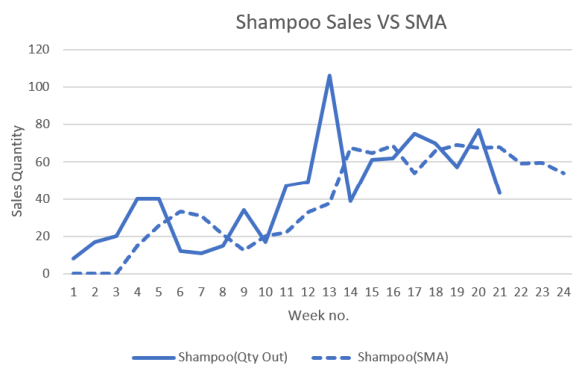


Figure 3.17: Shampoo Sales VS SMA

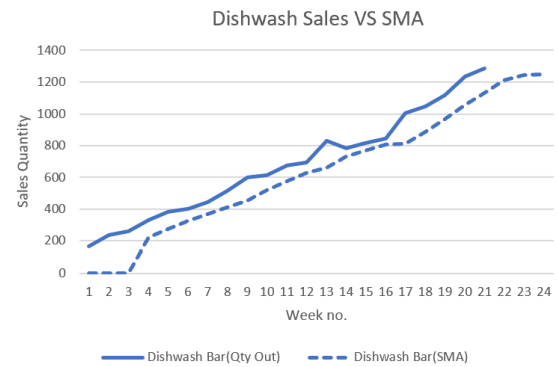


Figure 3.18: Dishwash Sales VS SMA

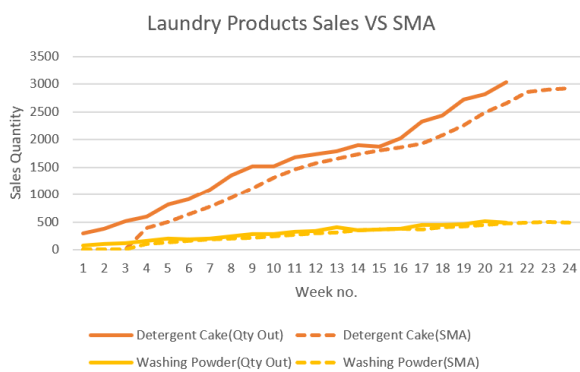


Figure 3.19: Laundry Products Sales VS SMA

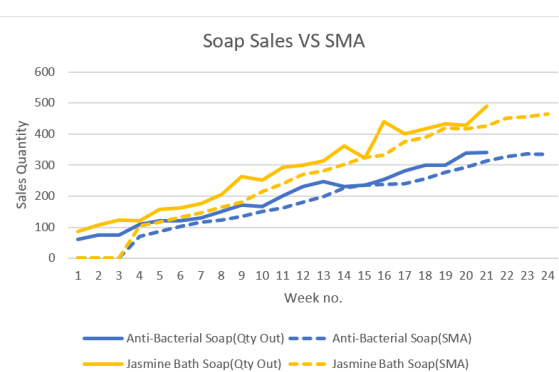


Figure 3.20: Soap Sales VS SMA

The line graphs for personal care and home care products over 24 weeks illustrate clear upward trends across all categories, with SMA lines closely tracking actual sales quantities. Herbal Toothpaste shows moderate, steady growth, while All Rounder Toothpaste stands out with much higher volumes and a robust upward trajectory. Multi-Purpose Cream and both Anti-Bacterial and Jasmine Bath Soaps display consistent, steady increases in demand, as reflected by their respective SMA lines. Shampoo sales rise significantly, featuring a sharp peak around week 13, but the overall trend remains upward. Dishwash Bar and Detergent Cake sales both show strong, consistent growth, with their SMA lines confirming this positive trajectory. Washing Powder sales are more stable, exhibiting a slight upward trend. Overall, the graphs confirm sustained and increasing demand across personal care and home care products, with SMAs validating the positive sales trend throughout the period.

Demand Forecasting for Week 22-24 for the products of Apparel and Footwear categories:

Week	Apparel and Footwear					
	White Dhoti (Qty Out)	White Dhoti(SMA)	Signature Cloth(Qty Out)	Signature Cloth(SMA)	Legging(Qty Out)	Legging(SMA)
1	4	-	3	-	3	-
2	2	-	1	-	1	-
3	2	-	1	-	1	-
4	3	3	4	2	1	2
5	6	2	1	2	4	1
6	1	4	2	2	0	2
7	5	3	2	2	8	2
8	5	4	7	2	2	4
9	8	4	0	4	3	3
10	9	6	4	3	2	4
11	5	7	5	4	7	2
12	13	7	5	3	6	4
13	14	9	6	5	3	5
14	6	11	3	5	0	5
15	7	11	4	5	9	3
16	12	9	3	4	18	4
17	8	8	5	3	13	9
18	11	9	5	4	13	13
19	8	10	5	4	18	15
20	6	9	3	5	17	15
21	10	8	1	4	13	16
22		8		3		16
23		8		2		15
24		9		2		15

Table 3.7: Actual Sales and Calculated SMA of different products of Apparel and Footwear categories

Wmn Belly(Qty Out)	Wmn Belly(SMA)	Wmn Chappal(Qty Out)	Wmn Chappal(SMA)	Mn Formal Shoes(Qty Out)	Mn Formal Shoes(SMA)
1	-	2	-	1	-
0	-	4	-	0	-
1	-	1	-	1	-
0	1	5	2	1	1
3	0	1	3	0	1
1	1	2	2	0	1
0	1	1	3	1	0
2	1	1	1	0	0
6	1	5	1	1	0
2	3	7	2	1	1
11	3	9	4	8	1
9	6	5	7	3	3
3	7	7	7	2	4
1	8	0	7	0	4
2	4	8	4	2	2
5	2	17	5	4	1
7	3	8	8	4	2
7	5	8	11	3	3
5	6	10	11	3	4
7	6	12	9	2	3
6	6	8	10	2	3
	6		10		2
	6		10		2
	6		9		2

Table 3.8: Actual Sales and Calculated SMA of different products of Apparel and Footwear categories

Line Charts of Actual Sales vs. Simple Moving Average for Apparel and Footwear Categories of Products:

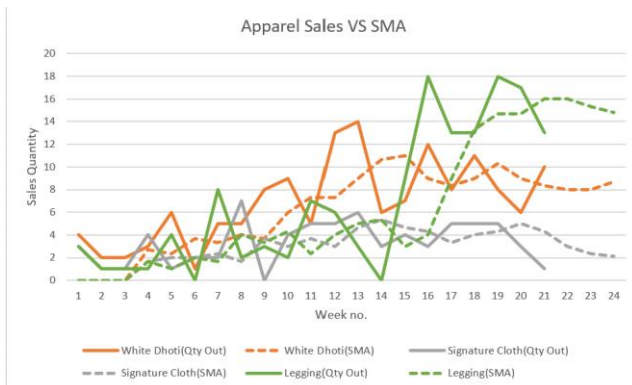


Figure 3.21: Apparel Actual Sales VS SMA

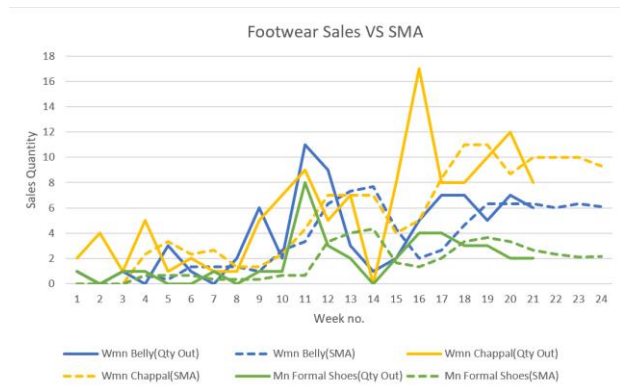


Figure 3.22: Footwear Actual Sales VS SMA

White Dhoti and Legging sales display rising trends, with some fluctuations, while Signature Cloth sales remain relatively stable at lower volumes. The SMA lines confirm these patterns. Women's Belly and Women's Chappal sales show growth with occasional peaks, while Men's Formal Shoes sales rise gradually. The SMA lines smooth out these fluctuations, indicating overall upward trends.

Inventory Optimization:

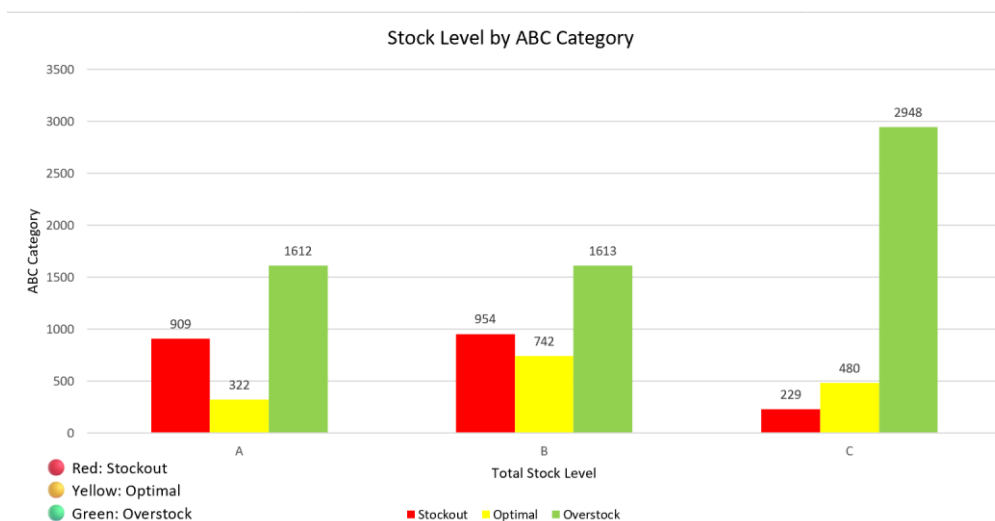


Figure 3.23: Bar Chart of Stock Level by ABC Category

The analysis reveals that overstock is the most significant inventory issue across all categories, as highlighted by both the data and the accompanying bar chart in figure 3.23. In Category A, representing the top 80% of value, overstocked units (1,612) substantially exceed both stockout (909) and optimal (322) levels. Similarly, in Category B, the next 15% of value, overstock (1,613 units) is

higher than both stockout (954) and optimal (742) units. The problem is most pronounced in Category C, the bottom 5% of value, where overstocked units soar to 2,948, dramatically surpassing stockout (229) and optimal (480) levels. This pattern underscores that excess inventory is a pervasive concern, particularly for lower-value items, while optimal stock levels remain consistently lower than both overstock and stockout figures in all categories.

Product Profitability Analysis:



Figure 3.24: Bar Chart of Top Ten Products by Total Profit

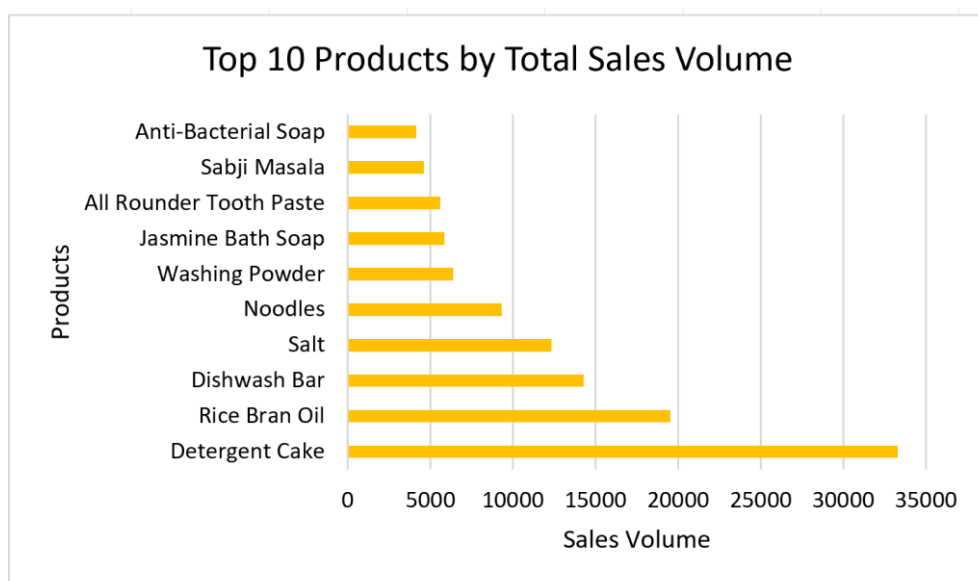


Figure 3.25: Bar Chart of Top Ten Products by Total Sales Volume

Top 10 Products by Total Profit:

Rice Bran Oil stands out with the highest total profit of ₹709,866, significantly ahead of all other products. It is followed by high-margin health supplements which are Nutricharge Woman, Nutricharge Man, and Nutricharge Veg Omega. Among grocery items, the most profitable are Atta, Turmeric Powder, Sabji Masala, and Salt. Additionally, products like Detergent Cake and Washing Powder also make it into the top 10 by profit.

Top 10 Products by Total Sales Volume:

Detergent Cake leads with the highest sales volume of approximately 33,300 units. Rice Bran Oil follows with strong volume. Other high-volume products include Dishwash Bar, Salt, Noodles, Washing Powder, Jasmine Bath Soap, All Rounder Tooth Paste, Sabji Masala, and Anti-Bacterial Soap.

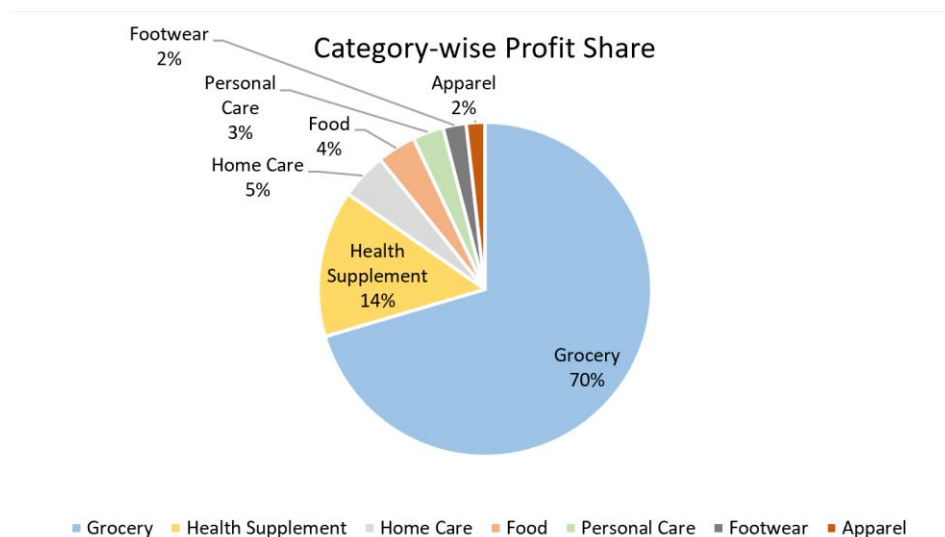


Figure 3.26: Pie Chart representing Category-wise Profit Share

Category-wise Profit Share:

Grocery contributes a dominant 70% of the total profit. Health Supplements are second with 14%, though with fewer items, indicating higher margins per item. Remaining categories contribute marginally as Home Care(5%), Food(4%), Personal Care(3%), Apparel and Footwear(2%) each.

Product Name	Total Profit	Total Sales Volume
Masoor Malka	3393	1097
Tea	1717	2385
Chilli Powder	2686	746
Coriander Powder	1484	1054
Turmeric Powder	20001	3049
Garam Masala	9074	1408
Sabji Masala	17618	4588
Atta	31893	1449
Rice Bran Oil	709866	19534
Noodles	14890	9306
Fiber Rich Biscuit	4965	1655
Jeera Biscuit	4253	2170
Mixed Pickle	5433	679
Ketchup	1627	1017
Gooddot Proteiz	10779	1773
Nutricharge Woman	83471	2386
Nutricharge Veg Omega	33075	1292
Nutricharge Man	49159	1707
Mixture	5372	920
Bhujia	1876	2606
Salt	16793	12348
Herbal Tooth Paste	1866	1296
Multi Purpose Cream	1801	2502
Shampoo	4608	900
All Rounder Tooth Paste	4018	5580
Detergent Cake	23978	33303
Dishwash Bar	10288	14289
Anti-Bacterial Soap	10588	4136
Jasmine Bath Soap	12166	5849
Washing Powder	18333	6366
White Dhoti	10005	145
Signature Cloth	4256	70
Legging	6714	142
Wmn Belly	15604	79
Wmn Chappal	1456	121
Mn Formal Shoes	9037	39

Figure 3.27: Heatmap highlighting Product-wise Total Profit and Total Sales Volume

Heatmap Analysis:

High Profit + High Volume:

Rice Bran Oil and Detergent Cake are standout performers.

Low Volume + High Profit:

Nutricharge Woman, Nutricharge Man, and Nutricharge Veg Omega indicate high value pricing and high profit per unit.

High Volume + Low Profit:

Products like Washing Powder, Salt, Dishwash Bar, and All Rounder Toothpaste indicate volume-driven sales but thin margins.

Low Volume + Low Profit:

Items such as Women Chappal, Women Belly, White Dhoti, and Signature Cloth appear to be low-performing products.

Please use the below link to see the original received data, inventory analysis and predicted data:

<https://drive.google.com/drive/folders/11UC9WaqDuw6QDZoJDCGNWnVPDIilTTAr?usp=sharing>

4 Recommendations

Tailored Inventory Strategies: Manage stable demand products with fixed reorder points and minimal safety stock, reducing the risk of both overstock and stockouts. Moderate and High fluctuation products require dynamic inventory management. So, for those products, regularly review and adjust reorder points and safety stock in response to changing demand based on recent sales trends and forecasts.

Prioritize High-Value Items: Focus inventory management efforts on Category A items to ensure optimal stock levels, minimizing both overstock and stockouts for these high-impact products.

Reduce Overstock, Especially for Low-Value Items: Implement stricter inventory controls for Category C items to prevent excessive overstock. Consider reducing order quantities, running promotions, or discontinuing slow-moving items to free up capital and storage space.

Maintain Using SMA for Short-Term Forecasting: Continue using the Simple Moving Average for short-term demand forecasting, as it provides actionable insights and helps align inventory with expected sales.

Promote High-Margin, High-Volume Products: Increase marketing and shelf space for top-performing products like Rice Bran Oil and Detergent Cake to maximize both profit and sales volume.

Review Low-Performing Products: Consider discontinuing low-sales-volume, low-profit products to focus resources on more profitable lines.

Optimize Pricing and Promotions: Adjust pricing strategies for products for high-demand, high-margin items, and consider small price increases if market conditions allow. For slow-moving items, use targeted promotions to boost turnover.